

Robotics, Lab assignment 1

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Due on: 9-10-2026, 23:59:59

To be addressed in groups of up to 3 students.

Introduction

The objective of the lab is to (i) raise awareness for the importance of studying robot dynamics, and (ii) to introduce realistic robotics simulation principles.

Consider a Golf player in the initial *swing* movement. The movement involves feet, legs, upper body, and a characteristic arm swing to make the club hit the ball (such that it reaches as close as possible to a given target (usually a hole in a green)). Figure 1 shows the chained sequence of movements that forms the *swing*.

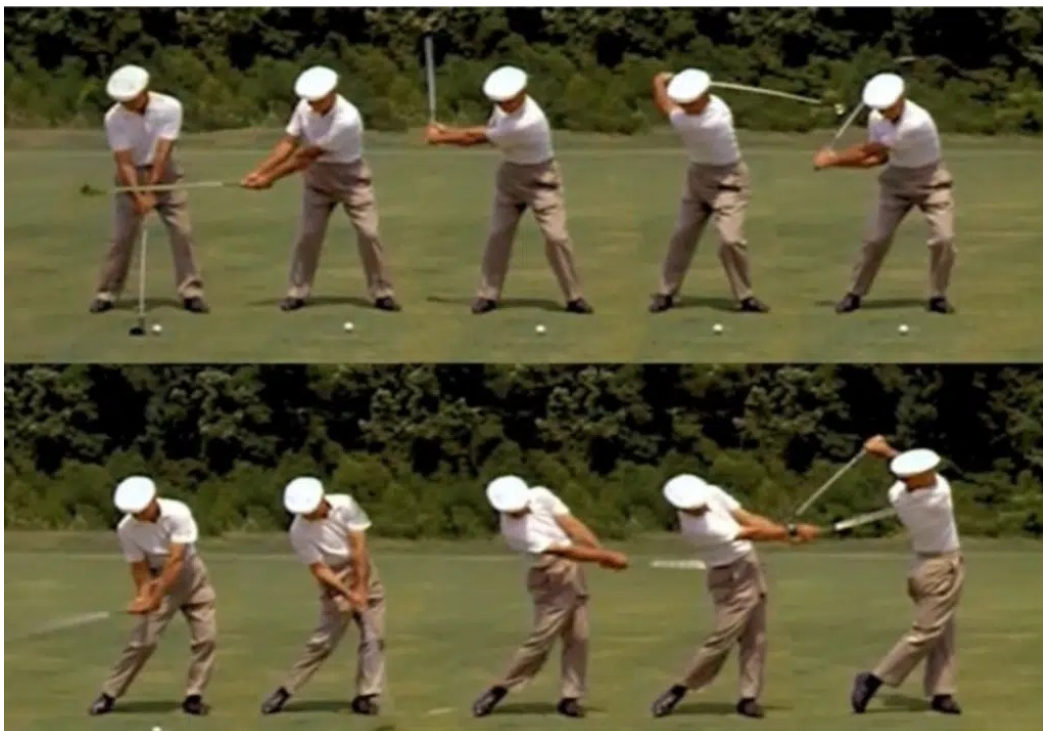


Figure 1 – Chained sequence of movements in a swing (adapted from <https://milolinesgolf.com/perfect-your-slow-motion-golf-swing-to-improve-faster/>).

The swing movement, the shape of the club, the speed at which the player's body moves, all of them are critical to transmit to the ball the adequate force/torque.

In this lab a simulation of the Golf swing movement is to be developed. In terms of Robotics, we'll consider the player and the club as the robot.

The simulation should be developed in MuJoCo (MUlti-JOint dynamics with COntact) + Python (this is software freely available online, for both Windows and Linux systems). The groups should address all of the following tasks and the solution for each of them presented separated.

Deliverables include (i) any written answers to the tasks below, and (ii) Python software, MuJoCo models developed, and instructions on how to run them.

Tasks

Task 1 – Develop a MuJoCo XML model for the ensemble Golf player+club.

List any simplifications/assumptions made and their potential impact in the realism of the simulation.

In your answer/solution, ensure that the XML MuJoCo model is duly commented and any additional files, e.g., STL files which allow for realistic renderings, are provided.

Task 2 – Design a set of tests that can be run under either the Simulate App in the MuJoCo software or using the Python environment, to show how realistic the model is.

Run the simulation on these tests. Consider the existence of disturbances in each degree-of-freedom (dof), e.g., the wrist of the player being shaky.

Task 3 – Write a short (one A4 page max) comment on the results obtained. Emphasize the good points of your results but do not hide the rest!

Solution hints

1. Initially, before putting the full model up for simulation, ensure that you master the basics of MuJoCo, e.g., implement a simple pendulum and a simple pendulum hitting a ball and ensure that they behave realistically.

If this is an exercise apparently unrelated with the goal of this lab, that is only apparent!

2. Sketch the model in a paper sheet.

3. Clearly identify the reference frames associated with the world and each link/rigid body that form the player and club.

4. Clearly identify the dof in the system. Some simplifications are admissible here, e.g., replace the biped player by a simpler structure, without the bipedal characteristics, or simplify the shape of the club (which, in a fully realistic simulation could be very complex – as the figure above illustrates).

5. Define the physical and geometric parameters of each rigid body, i.e., masses and inertias.

6. Define the friction properties at each dof.

7. Define the level of actuation for each dof, e.g., limits of the actuator(s).

There is a vast collection of online resources on MuJoCo. For example:

<https://mujoco.readthedocs.io/en/latest/overview.html>

<https://mujoco.readthedocs.io/en/latest/python.html>

<https://mujoco.readthedocs.io/en/latest/modeling.html>

<https://github.com/google-deepmind/mujoco#getting-started>